Answers

Worksheet

Volume of prisms

Task A: Calculating the volume of any prism

1) Sketch all the cuboids with integer lengths, that have a volume of 18 cm^3

There are four cuboids with integer lengths $1 \times 1 \times 18$

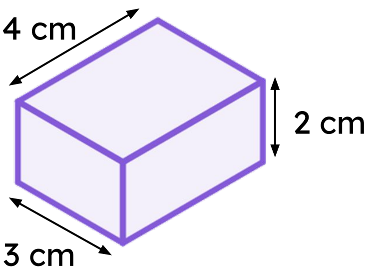
$1 \times 2 \times 9$

$1 \times 3 \times 6$

$2 \times 3 \times 3$

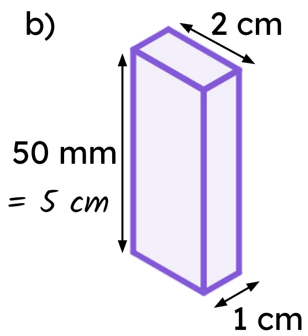
2) Calculate the volume of these cuboids

a)



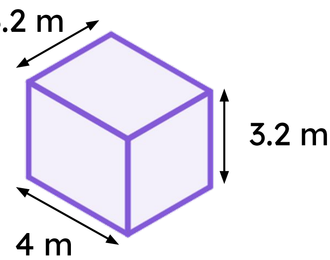
$3 \times 2 \times 4 = 24$ cm^3

b)



$5 \times 2 \times 1 = 10$ cm^3

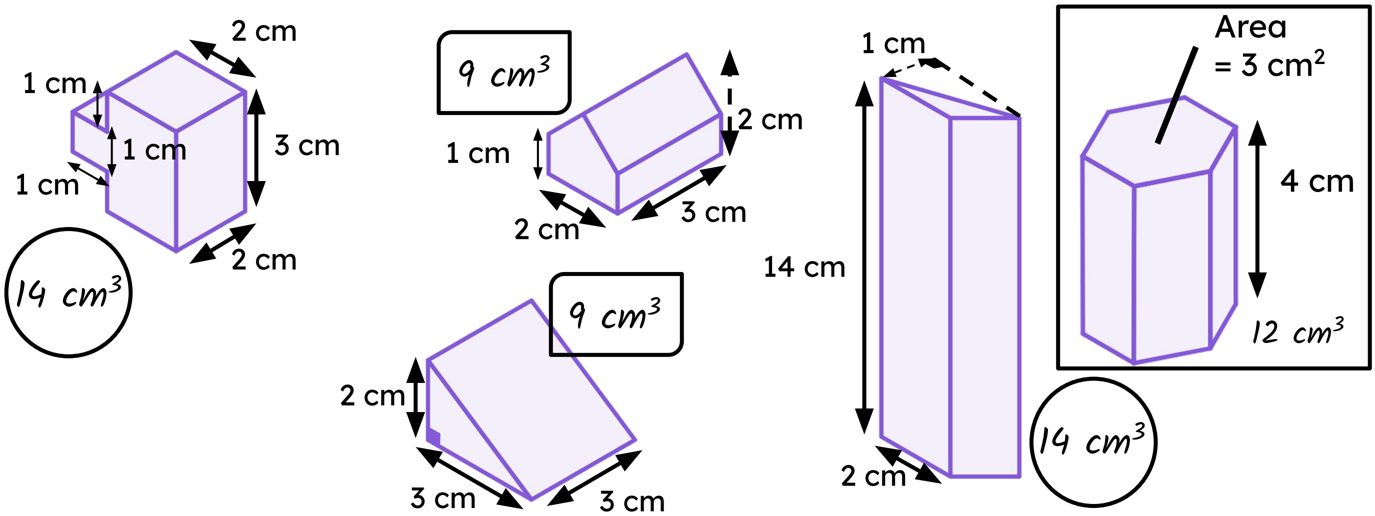
c)



$4 \times 3.2 \times 3.2$

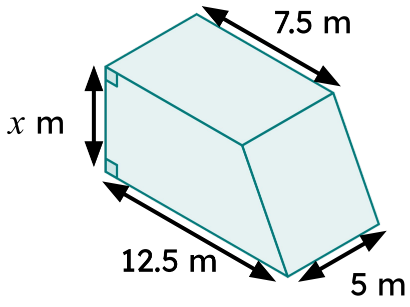
$= 40.96$ m^3

3) There are pairs of shapes with the same volume. Which shape does not have a pair?



Task B: Using the volume of a prism

1) Complete the missing parts of the calculation.



Volume = 300 m³

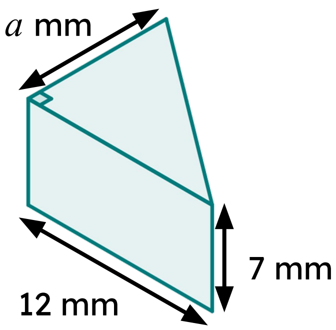
$$\left(\frac{1}{2} \times (12.5 + 7.5\ ) \times x\right) \times 5 = 300$$

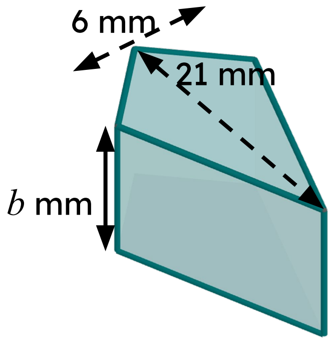
$$\left(\frac{1}{2} \times 20 \times x\right) \times 5 = 300$$

$$10x = 60$$

$$x = 6$$

2) Both prisms have a volume of 378 mm³. Work out the missing length in each prism.





$$\left(\frac{1}{2} \times 12 \times a\right) \times 7 = 378$$

$$6a \times 7 = 378$$

$$42a = 378$$

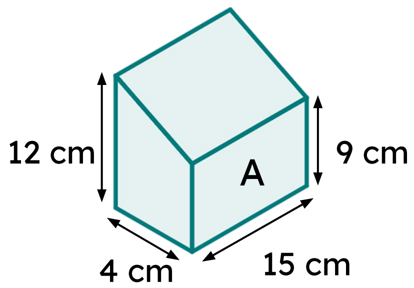
$$a = 9$$

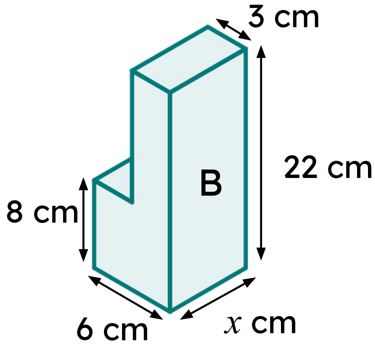
$$21 \times 6 \times b = 378$$

$$126b = 378$$

$$b = 3$$

3) Prism A and prism B have the same volume. Work out the missing length of prism B.





Volume of prism A = volume of prism B

$$\left(\frac{1}{2} \times (12 + 9) \times 4\right) \times 15 = (8 \times 6 + 14 \times 3) \times x$$

$$630 = 90x$$

$$7 = x$$